

Performance Testing Phase

Project Performance Test

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether it's through targeted promotional campaigns, adjustments in product formulations, or personalized

recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

5.1 Performance Testing Results

Test #	Description	Expected Result	Actual Result	Status
PT-01	Load cosmetics dataset into Tableau	All 11 fields recognized with correct types	1,472 rows loaded, 11 fields — no errors	PASS
PT-02	Pivot skin-type columns	Row count expands to 7,360 with Skin_Type / Suitable created	Confirmed: 7,360 rows, both fields present	PASS
PT-03	Category filter click response	Other charts filter within 2 seconds	Filtered in under 1 second	PASS
PT-04	Brand filter search	Top Brands by Category updates instantly	Verified across multiple brand selections	PASS
PT-05	Scatter plot rendering	~1,450 marks render without lag	Rendered smoothly, no overlap errors	PASS
PT-06	Cross-browser compatibility	Loads and filters work in all browsers	Verified in Chrome, Edge, and Safari	PASS
PT-07	Published link accessibility	Loads without requiring login	Tableau Public URL loads directly	PASS
PT-08	Filter reset	Deselecting restores full dataset across all charts	Confirmed on repeated tests	PASS

5.2 Performance Summary

Metric	Value
Total Dataset Rows (post-pivot)	7,360
Total Columns (post-pivot)	9
Number of Derived Fields	2 (Skin_Type, Suitable)
Number of Filters Used	2 (Category click-filter action, Brand filter)
Number of Visualizations / Charts	4
Number of Dashboards	1
Overall Performance Rating	PASS — No performance issues detected

All eight test cases passed with no defects logged. Because the underlying dataset is modest in size, filter and cross-chart actions responded well within the 2-second target, confirming the dashboard does not require data extraction or aggregation to remain responsive on Tableau Public's free hosting tier.